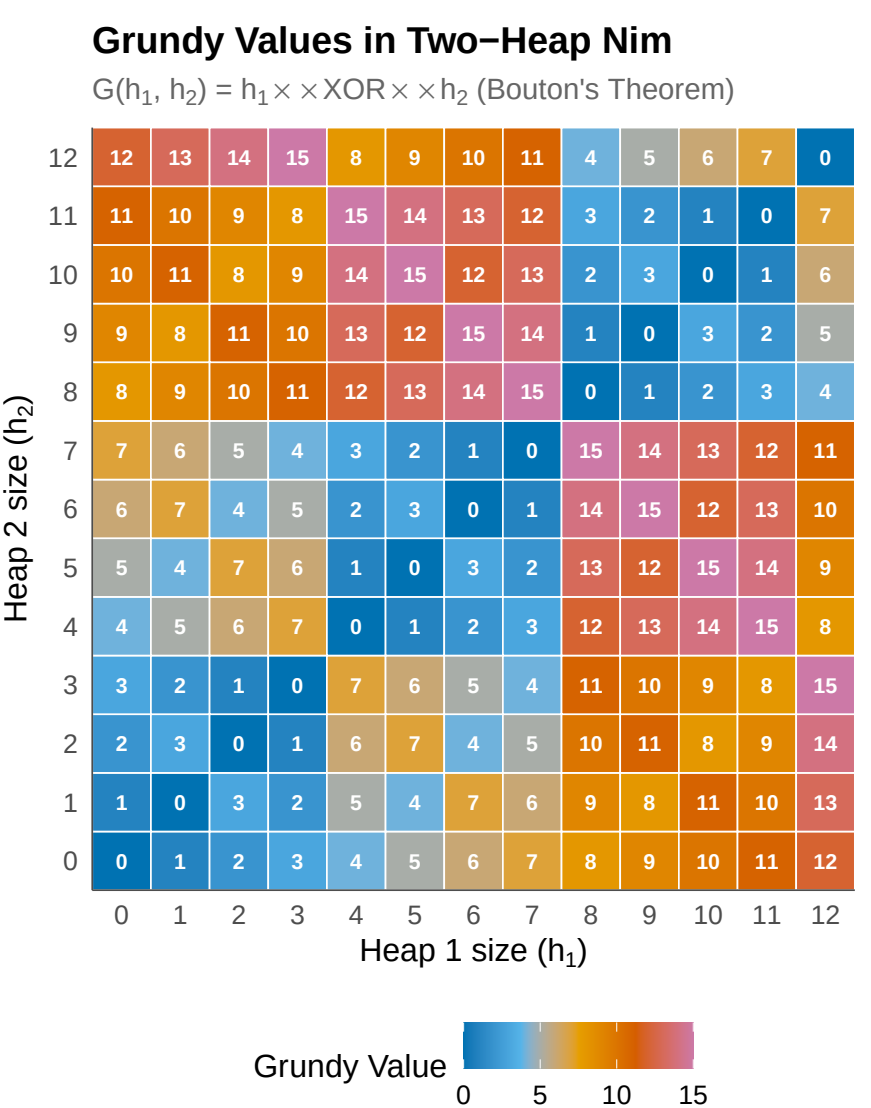
Grundy Values in Two−Heap Nim
G(h1, h2) = h1 × ×XOR× × h2 (Bouton's Theorem)
Heap 2 size (h2)
12 12 13 14 15 8 9 10 11 4 5 6 7 0
11 11 10 9 8 15 14 13 12 3 2 1 0 7
10 10 11 8 9 14 15 12 13 2 3 0 1 6
9 9 8 11 10 13 12 15 14 1 0 3 2 5
8 8 9 10 11 12 13 14 15 0 1 2 3 4
7 7 6 5 4 3 2 1 0 15 14 13 12 11
6 6 7 4 5 2 3 0 1 14 15 12 13 10
5 5 4 7 6 1 0 3 2 13 12 15 14 9
4 4 5 6 7 0 1 2 3 12 13 14 15 8
3 3 2 1 0 7 6 5 4 11 10 9 8 15
2 2 3 0 1 6 7 4 5 10 11 8 9 14
1 1 0 3 2 5 4 7 6 9 8 11 10 13
0 0 1 2 3 4 5 6 7 8 9 10 11 12
0 1 2 3 4 5 6 7 8 9 10 11 12
Heap 1 size (h1)
Grundy Value
0 5 10 15